Near-Real-Time Satellite Tracking, Orbital Dynamics, and Relativistic Time Dilation

Final engineering report
MATLAB R2026b
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Abstract

This project implements a reproducible MATLAB application for real-record satellite tracking, orbital-model comparison, and relativistic clock analysis. Six spacecraft spanning low, medium, and geosynchronous Earth orbit were propagated with the Vallado/CelesTrak SGP4 implementation and compared with a custom ode45 point-mass two-body model over 1, 6, and 24 hours. Position differences were transformed to radial, along-track, and cross-track coordinates. A weak-field clock model produced +38.452079 microseconds/day for a theoretical GPS orbit relative to an idealized stationary equatorial surface clock. Final verification completed 120/120 simulations and passed 17/17 consolidated checks.

Executive results

Metric	Measured result
Real spacecraft / orbital regimes	6 / LEO, MEO, GEO
Simulation campaign	120/120 passed
Final verification	17/17 passed
Maximum two-body - SGP4 separation	1,781.797863 km
Maximum relative energy drift	6.0711e-10
GPS net relativistic correction	+38.452079 us/day

Interpretation rule: two-body - SGP4 separation is model disagreement, not independently validated tracking error.

1. Objectives and application architecture

The engineering objective was to connect real orbital records to physics, numerical analysis, verification evidence, and an interactive display. The completed catalog contains ISS (ZARYA), NAVSTAR 48, NAVSTAR 51, NOAA-20, Landsat 9, and GOES-16. The acceptance run uses cached records from 18-19 October 2026 for reproducibility; the cache is evidence, not a current operational ephemeris.

Data and computation flow

1. Ingest	2. Propagate	3. Compare	4. Present
CelesTrak OMM/TLE validation and cache	SGP4 TEME states + ode45 two-body	J2000/FK5 and RTN metrics	CSV/MAT, figures, dashboard, 3-D globe

Software organization

Module	Responsibility
src/io	Provider queries, validation, cache provenance, EOP loading
src/propagation	Vallado SGP4 catalog propagation and ode45 interface
src/frames	TEME to J2000/FK5 conversion
src/analysis	Orbital elements, RTN, diagnostics, campaign metrics
src/relativity	Kinematic, gravitational, and combined clock models
src/dashboard + apps	Verified-state selection, plotting, and export
tests + scripts	Reproducible workflows and acceptance checks

The dashboard reads the same saved SGP4/J2000 arrays and Day 9 timing results used by the analysis. It does not substitute display-only trajectories. A separate control opens the native satelliteScenario globe when Satellite Communications Toolbox is installed.

2. Mathematical foundations

Two-body dynamics

The numerical model integrates r_ddot = -mu r / |r|^3 in a consistent kilometre-second unit system. Independent circular benchmarks are v_c = sqrt(mu/r) and T = 2 pi sqrt(r^3/mu). Elliptical apsides are r_p = a(1-e) and r_a = a(1+e). Specific energy epsilon = v^2/2 - mu/r and specific angular momentum h = r x v are invariants of the continuous point-mass model and therefore support numerical verification.

SGP4 and frame discipline

SGP4 returns states in the True Equator, Mean Equinox frame. Direct subtraction from an arbitrary ECI state would mix frame effects with dynamics. Each SGP4 history is therefore converted through the documented Vallado IAU-76/FK5 path using CelesTrak Earth-orientation data. The first converted J2000/FK5 state initializes ode45. The difference is defined as two-body minus SGP4 and projected into the SGP4 radial, along-track, and cross-track basis.

Relativistic timing

The sign convention is satellite proper time minus reference-clock proper time. The reference is an idealized stationary clock at the WGS-84 equatorial radius in a spherical, nonrotating Earth model. The leading rates are:

gravitational: delta rate = (mu/c^2)(1/R_E - 1/r)
kinematic: delta rate = -v^2/(2c^2)

Rates are integrated with respect to elapsed coordinate time and reported in microseconds per day. The reference convention is essential: the familiar GPS +38.5 microseconds/day value is not independent of the chosen clock and coordinate definition.

Model boundaries

Included	Excluded
SGP4 mean-element propagation	Independent precision truth ephemeris
Point-mass two-body ode45 model	J2, drag, third bodies, SRP, maneuvers
TEME to J2000/FK5 conversion	Full IAU 2006/2000A reduction
First-order gravity + special relativity	Rotation, tides, signal path, GPS system time

3. Verification and numerical campaign

The deterministic campaign spans five semimajor-axis regimes, four eccentricities, and six inclinations for 120 full-orbit cases. Each case contains 181 requested output samples. A case passes only when all states are finite, relative energy and angular-momentum drift remain below 1e-7, and numerical perigee and apogee agree with the analytical values within 0.25 km.

Verification item	Measured / criterion
Automated orbit simulations	120 passed / at least 100
Circular-period closure	2.645e-10 km / 0.01 km
Maximum relative energy drift	6.0711e-10 / 1e-7
Maximum relative angular-momentum drift	3.0380e-10 / 1e-7
TEME-J2000 rotation orthogonality	5.088e-16 / 1e-10
RTN reconstruction closure	9.444e-13 km / 1e-9 km
Tightest 6-hour convergence difference	0.0001157 m / 10 m
Consolidated acceptance checks	17 passed, 0 failed, 0 skipped

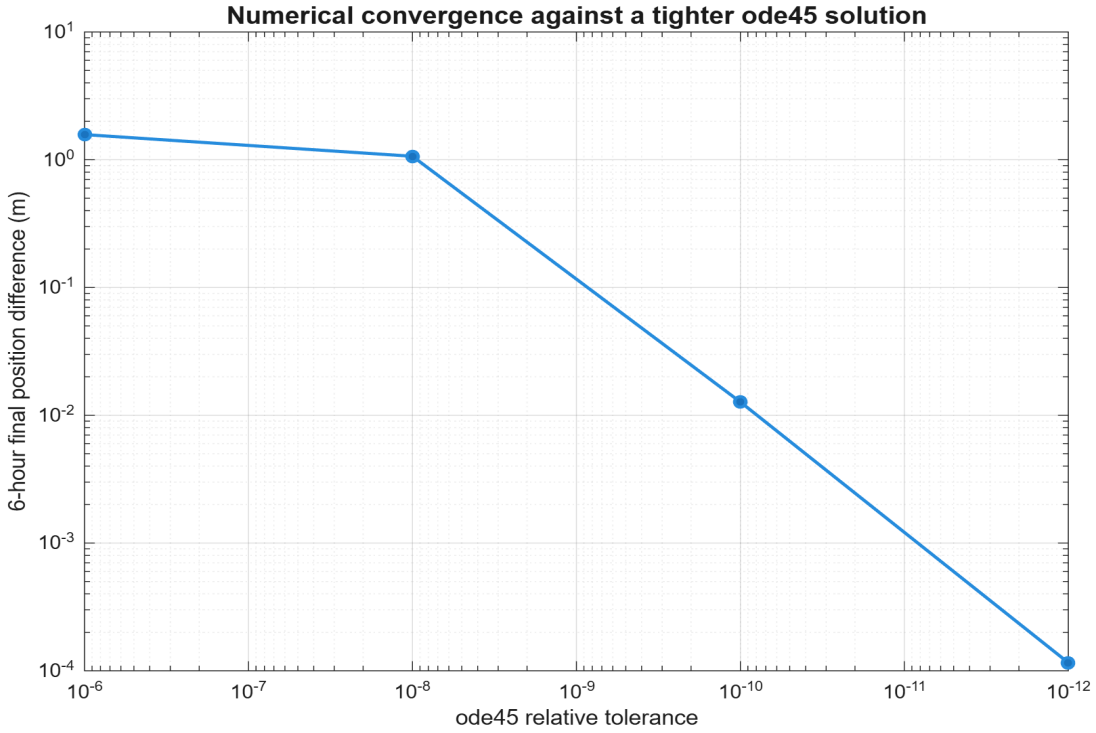


Figure 1. ode45 numerical convergence against a tighter two-body solution. This demonstrates numerical convergence of the implementation, not real-world orbit prediction accuracy.

4. Orbital-model comparison results

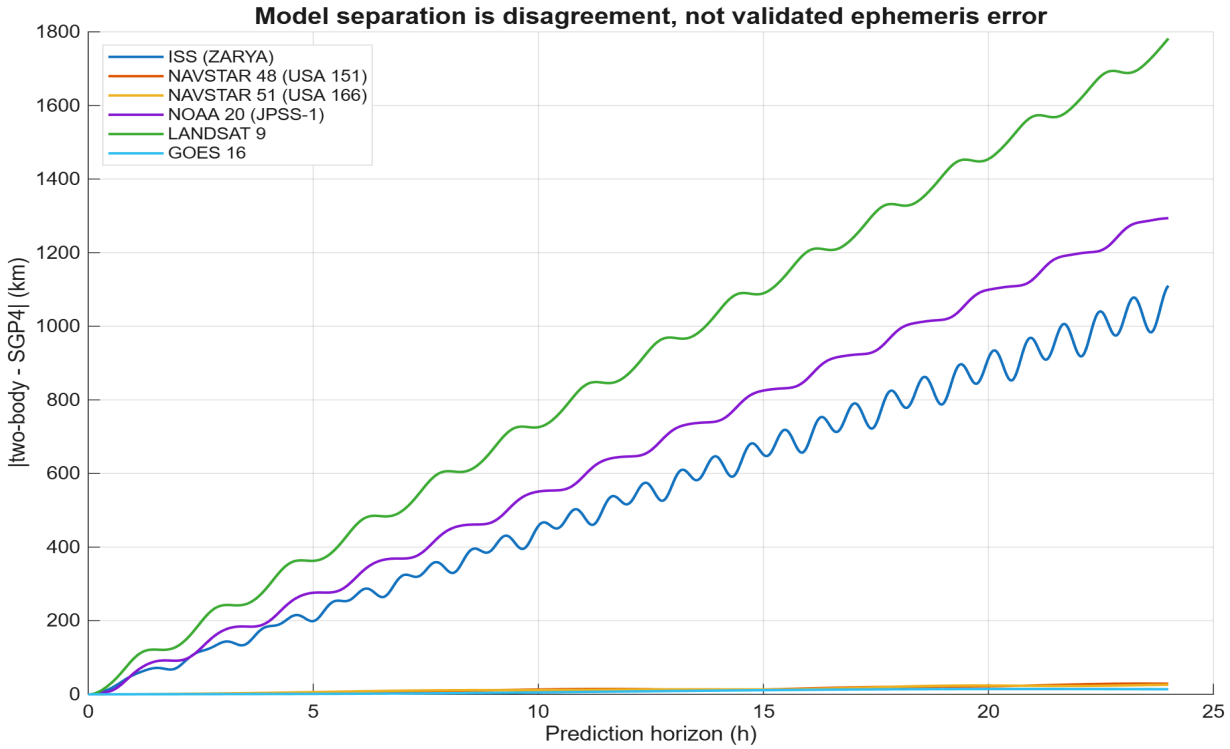


Figure 2. Magnitude of two-body minus SGP4 position separation for six real records.

Prediction horizon	Maximum separation	Mean case RMS
1 hour	95.91 km	16.31 km
6 hours	466.12 km	100.78 km
24 hours	1,781.80 km	404.87 km

The global RMS separation across all stored samples is 575.823254 km. Landsat 9 produced the largest 24-hour separation, with a 1,765.071 km maximum absolute along-track component and a 240.977 km radial component. The trend is expected: the two-body model omits drag, oblateness, higher geopotential terms, lunisolar gravity, solar radiation pressure, and maneuver representation. LEO phase is particularly sensitive to small mean-motion differences. No independent truth ephemeris was used, so the result must remain labeled model disagreement.

5. Relativistic timing and application performance

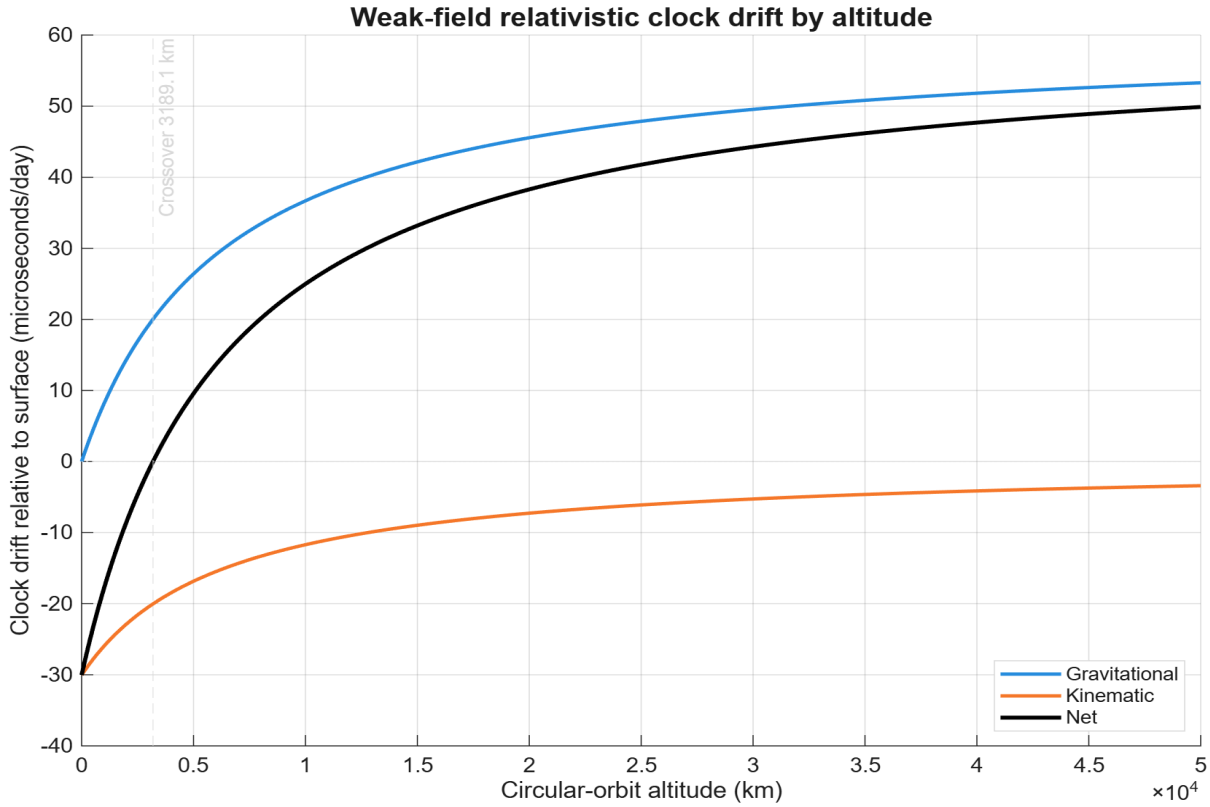


Figure 3. Gravitational, kinematic, and net clock drift for idealized circular orbits relative to a stationary equatorial surface clock.

Timing / application metric	Measured result
GPS gravitational contribution	+45.660757 us/day
GPS kinematic contribution	-7.208677 us/day
GPS net contribution	+38.452079 us/day
Difference from +38.5 benchmark	0.047921 us/day (0.1245%)
Idealized zero-drift crossover	3,189.069 km altitude
Dashboard snapshot update latency	0.884 ms mean over 100 calls
Cold hidden-UI startup	20.166 s on acceptance machine

ISS is kinematically dominated and accumulates approximately -24.48 us/day in this convention. Real GPS records are near +38.44 us/day, while GOES-16 is approximately +46.45 us/day. Startup includes MATLAB graphics initialization and should not be generalized to warm sessions or different hardware.

6. Limitations, conclusions, and next steps

Acceptance limitations

- Cached element epochs are reproducible evidence, not current operational tracking data.

- SGP4 output is not a precision truth ephemeris, and no surveyed truth trajectory was used.

- The two-body model intentionally omits major perturbations and orbit-determination uncertainty.

- The clock model excludes Earth rotation, geopotential harmonics, tides, signal delays, and operational GPS time realization.

- A clean MATLAB installation without Satellite Communications Toolbox was unavailable; missing-toolbox behavior is not tested.

- Hidden dashboard startup and data integrity were automated; interactive close behavior was not manually tested in the headless run.

Conclusions

The project meets its central portfolio objectives: real orbital data, validated SGP4 execution, independent ode45 propagation, frame-consistent RTN analysis, special and gravitational relativity, more than 100 automated runs, more than 10 final checks, exported evidence, and an interactive dashboard. The verification supports numerical and software correctness within the tested scope without overstating model agreement as operational tracking accuracy.

Recommended extensions

Add explicit orbital-record age alarms; compare against an independent precision ephemeris; implement J2, drag, lunisolar gravity, and covariance propagation; add ground-station access and link analysis; and package the programmatic dashboard as an App Designer application after manual cross-platform testing.

References

1. Vallado, D. A., Fundamentals of Astrodynamics and Applications.

2. Vallado, Crawford, Hujsak, and Kelso, Revisiting Spacetrack Report #3, AIAA 2006-6753.

3. CelesTrak, General Perturbations data and SGP4 reference implementation.

4. MathWorks, MATLAB ode45, satelliteScenario, and Satellite Communications Toolbox documentation.

5. Ashby, N., Relativity in the Global Positioning System, Living Reviews in Relativity.